Sri Lanka Institute of Information Technology



IT3091 – Machine Learning

Final Report

Banking & Finance – Bank Marketing

Campaign Response Prediction and Marketing Profile Analysis

Group ID: 2026-DS-02

Lab Group: Y3.S1.WD.DS.0102

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1. Introduction and Project Overview

1.1 Business problem

A Portuguese retail bank sells term deposits through telephone marketing campaigns. Only about **one client in nine** subscribes, so most calls are wasted: they cost call-centre time and annoy clients who are not interested. The bank wants to use the information it already holds about its clients, their contact history and the economic situation to decide **who to call first**.

Stakeholder: the bank's marketing / campaign managers who plan the call list, and the call-centre team that carries it out.

1.2 Track, lenses and machine-learning task

Item	Choice
Track	Guided Data Track – Banking & Finance – Bank Marketing (UCI)
Primary lens	**Campaign Response Prediction** – estimate each client's likelihood of subscribing so the call list can be ranked before calling
Secondary lens	**Marketing Profile Analysis** – identify the client and campaign characteristics linked to subscribing, to explain the predictions and guide targeting
ML task	Supervised binary classification, evaluated as a ranking problem (PR-AUC)
Target	y – has the client subscribed to a term deposit? (yes = 1, no = 0)
Unit of analysis	One row = one client contact in a campaign (the last contact of that campaign)
Output	A score per client → ranked call list, a call / no-call rule, and customer profiles

1.3 Problem framing canvas

Element	Definition
Business scenario	A Portuguese retail bank runs telephone marketing campaigns to sell term deposits. Most calls do not lead to a subscription (≈11% success), so call-centre time is spent mostly on clients who say no.
Stakeholder	The bank's marketing / campaign managers who plan the call list, and the call-centre team that carries it out.
Decision need	*Before* a campaign starts: which clients should be contacted first, and which client/campaign characteristics are linked to a "yes"?
Primary lens	**Campaign Response Prediction:** estimate each client's likelihood of subscribing so the call list can be ranked.
Secondary lens	**Marketing Profile Analysis:** identify the characteristics associated with subscription (from EDA and model interpretation) to explain the predictions and guide campaign targeting. It supports the primary lens: prediction says *who*, profiling explains *why*.
Unit of analysis	One row = one client contacted in a campaign (last contact of that campaign), described by client, contact, campaign-history and economic-context attributes. To be confirmed in data understanding (Step 2).
ML task	Supervised binary classification.
Target	y: has the client subscribed to a term deposit? (yes = positive class, no).

Element	Definition
Output	A probability score per client → ranked call list; plus a yes/no decision at a business-chosen threshold, and a set of evidence-based response profiles.
Allowed inputs	Only information available **before** the call: client data, contact channel/timing, previous-campaign history, economic indicators. `duration` is excluded (known only after the call; see decision D3).
Success criteria	(1) Clearly outperforms the Logistic Regression baseline and the "always no" reference on PR-AUC and recall/F1 for the `yes` class; (2) useful ranking, e.g. the top-scored 20% of clients captures a much larger share of subscribers than 20%; (3) results are stable across cross-validation folds; (4) profiles are interpretable to non-technical managers.
Key constraints / risks	Strong class imbalance; missing values coded as `"unknown"`; potential leakage (`duration`); data from 2008–2010 in one country (may not generalise); sensitive attributes (age, marital status) raise fairness concerns.
Business value	Fewer wasted calls and lower campaign cost, more subscriptions per call, less customer annoyance from unwanted calls, and clearer understanding of which client segments respond.
What the model is NOT for	Automatically excluding customers from products, or making credit decisions; it only supports prioritisation of marketing calls, with human oversight.

1.4 Workflow

The project followed the 10-step workflow proposed in the initial report. Each step corresponds to a part of the final notebook and every important choice was recorded in the decision log (Section 10).

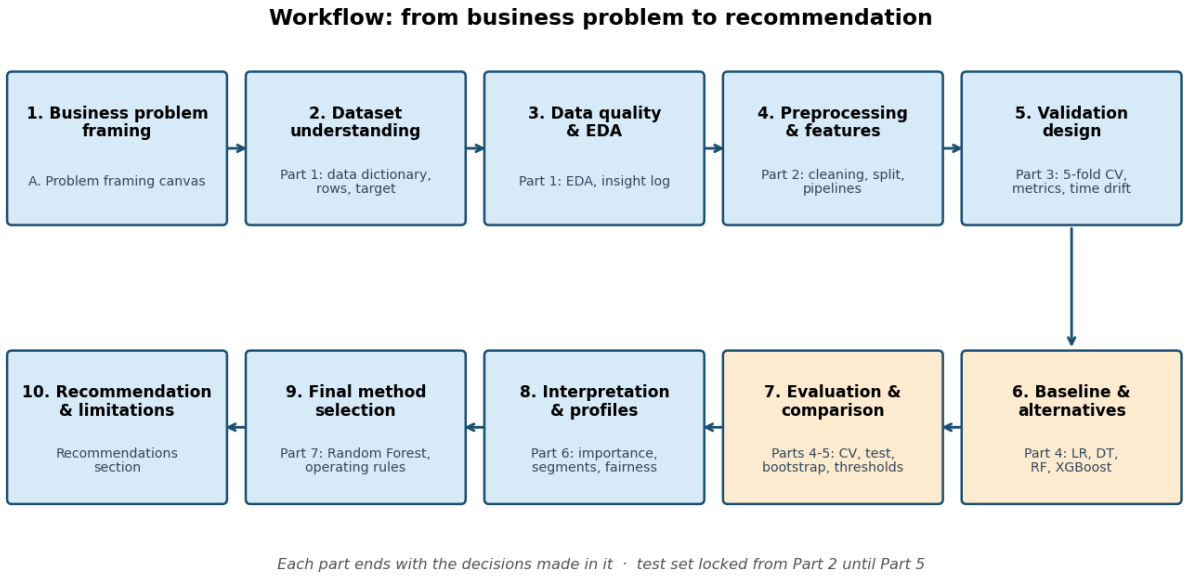


Figure 1 – Workflow from business problem to recommendation

1.5 Team and responsibilities

Each member owned one model and one main stage of the project (decision D30); all members reviewed every part.

Member	Student No	Model	Main stage	Decisions owned
Karunathilaka S.H.T.R	IT24103860	Logistic Regression (baseline)	Problem framing, data understanding & EDA	D1, D2, D4, D11, D32
Jayasuriya J.A.Y.N	IT24104178	Decision Tree	Preprocessing & features; segment rules; responsible-AI check	D3, D7–D10, D13–D15, D17, D25, D26
Thulmanthi W.A.S	IT24104110	Random Forest (final model)	Interpretation; final model selection; notebook integration	D6, D16, D19, D24, D27
Thennakoon T.M.S.D	IT24103869	XGBoost	Validation design; model comparison; test evaluation; recommendations	D5, D12, D18, D21, D22, D23, D28

2. Dataset and Data Dictionary

2.1 Dataset

Item	Value
Source	UCI Machine Learning Repository – Bank Marketing (https://archive.ics.uci.edu/dataset/222/bank+marketing); Moro, Cortez & Rita (2014)
File used	`bank-additional-full.csv` (separator `;`), stored in `data/raw/`
Size	41,188 rows × 21 columns (20 inputs + target); 41,176 rows after removing 12 duplicates
Period	May 2008 – November 2010; rows are ordered by date
Target balance	4,640 yes (11.27%) vs 36,548 no – imbalance 7.9 : 1
Missing values	No blank cells; missing categorical values are coded as "unknown" (26.0% of rows contain at least one)

Why this version (D1): the UCI download contains four files. `bank-additional-full.csv` was chosen because it contains five national economic indicators that are known before a call and improve prediction, and it has all rows. `bank-full.csv` (older, no economic indicators) and the two 10% samples were not used. The trade-off is that the account `balance` column is not available.

2.2 Data dictionary

Column	Group	Type	Description	Values / range	% unknown	Known before call?
age	Client	Numeric	Client age in years	17 – 98	0	Yes
job	Client	Nominal	Type of job (12 levels)	admin., blue-collar, … student, unknown	0.80	Yes
marital	Client	Nominal	Marital status (divorced includes widowed)	divorced, married, single, unknown	0.19	Yes
education	Client	Ordinal	Education level (8 levels)	basic.4y … university.degree, unknown	4.20	Yes
default	Client	Binary + unknown	Has credit in default?	no, unknown, yes	20.87	Yes
housing	Client	Binary + unknown	Has a housing loan?	no, unknown, yes	2.40	Yes
loan	Client	Binary + unknown	Has a personal loan?	no, unknown, yes	2.40	Yes
contact	Contact	Binary	Contact type	cellular, telephone	0	Yes (planned)
month	Contact	Nominal	Month of last contact	mar – dec (10 months)	0	Yes (planned)
day_of_week	Contact	Nominal	Weekday of last contact	mon – fri	0	Yes (planned)
duration	Contact	Numeric	Last call duration (seconds)	0 – 4,918	0	**No – leakage**
campaign	History	Count	Contacts in this campaign (incl. last)	1 – 56	0	Partly
pdays	History	Numeric + placeholder	Days since previous campaign contact (999 = never)	0 – 999	0	Yes
previous	History	Count	Contacts before this campaign	0 – 7	0	Yes
poutcome	History	Nominal	Outcome of the previous campaign	failure, nonexistent, success	0	Yes
emp.var.rate	Economic	Numeric	Employment variation rate (quarterly)	−3.4 – 1.4	0	Yes (public)
cons.price.idx	Economic	Numeric	Consumer price index (monthly)	92.201 – 94.767	0	Yes (public)
cons.conf.idx	Economic	Numeric	Consumer confidence index (monthly)	−50.8 – −26.9	0	Yes (public)
euribor3m	Economic	Numeric	Euribor 3-month rate (daily)	0.634 – 5.045	0	Yes (public)
nr.employed	Economic	Numeric	Number of employees, thousands (quarterly)	4,963.6 – 5,228.1	0	Yes (public)
y	Target	Binary	Subscribed to a term deposit?	no, yes	0	Target

3. Data Understanding and Exploratory Data Analysis

Exploratory analysis answered three questions: what the data looks like, which characteristics are linked to subscribing, and which data-quality issues must be handled before modelling.

3.1 Key findings

- **Strong imbalance:** only 11.27% subscribed. A model that always predicts "no" is 88.73% accurate but finds no subscribers, so accuracy cannot be the main metric.
- **Previous campaign success** is the strongest client signal: 65.11% of previously successful clients subscribe again (χ^2 = 4,230.5, the largest of all categorical features).
- **Economic context:** subscriptions are much higher when interest rates and employment are low (single-feature AUC 0.72–0.75 for `nr.employed`, `euribor3m`, `emp.var.rate`).
- **Timing and channel:** March, September, October and December reach 44–51% but May, the busiest month, only 6.43%; mobile contacts convert at 14.74% vs 5.23% for landlines.
- **Who:** students (31.43%) and retired clients (25.23%) respond best. Age is **U-shaped** (17–25: 20.95%, 41–50: 8.17%, 71+: 47.87%), a non-linear effect.
- **Contact fatigue:** 13.04% subscribe after one call but only 3.56% after 10 or more.

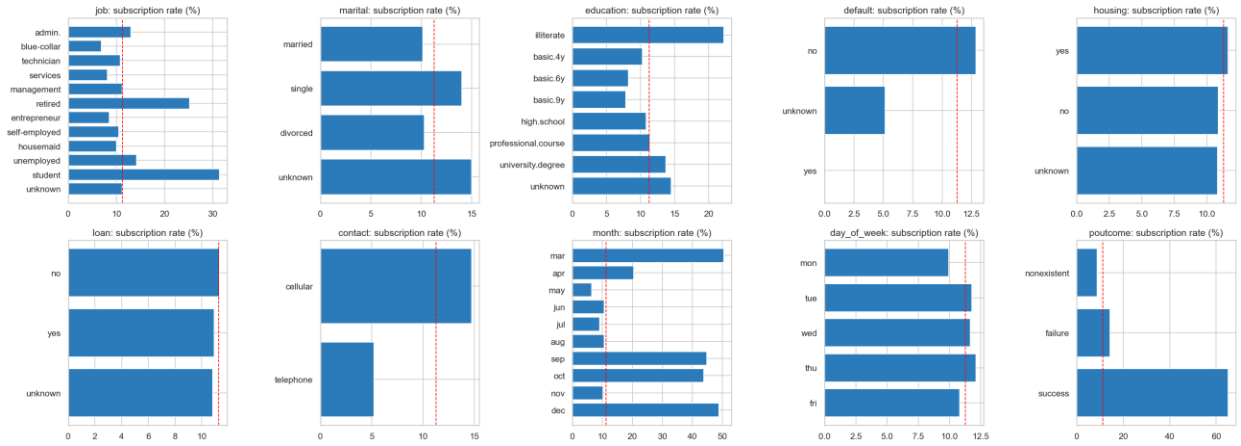


Figure 2 – Subscription rate per category (red line = overall 11.27%)

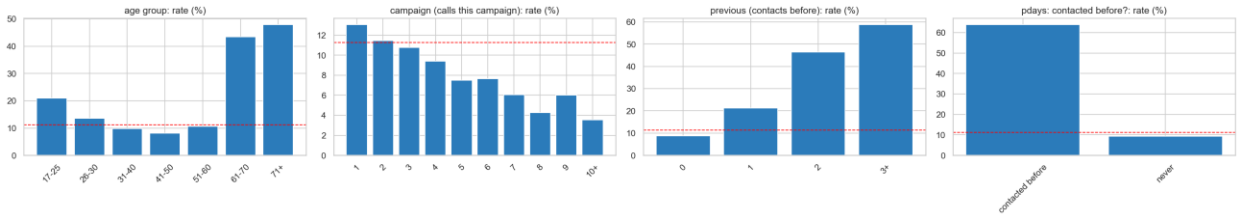


Figure 3 – Subscription rate by age group, calls, previous contacts and prior contact

3.2 Data quality and risks

- **"unknown" values** in 26.0% of rows are informative (unknown `default`: 5.15% yes vs 12.88%), so they are kept as a category.
- **12 exact duplicates** (no conflicting labels) and a **`pdays` placeholder** (999 in 96.3% of rows; 4,110 rows contradict its documented meaning).
- **Outliers are real, not errors** (ages 70–98, up to 56 calls) and are predictive (clients over 69.5 subscribe at 47.12%).
- **Multicollinearity:** `euribor3m`, `emp.var.rate` and `nr.employed` have VIF 64.3, 33.1 and 31.6 (Spearman 0.93–0.94).
- **Time drift:** the yes-rate rose from 4.84% (2008) to 19.48% (2009) and 52.14% (2010) as interest rates fell.
- **Leakage:** `duration` alone has AUC 0.818, higher than any pre-call input (Figure 5).

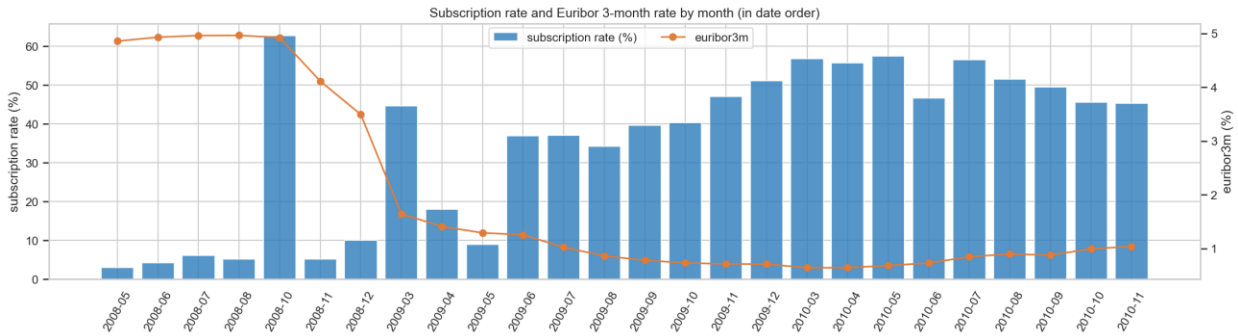


Figure 4 – Subscription rate and Euribor rate by month, in date order

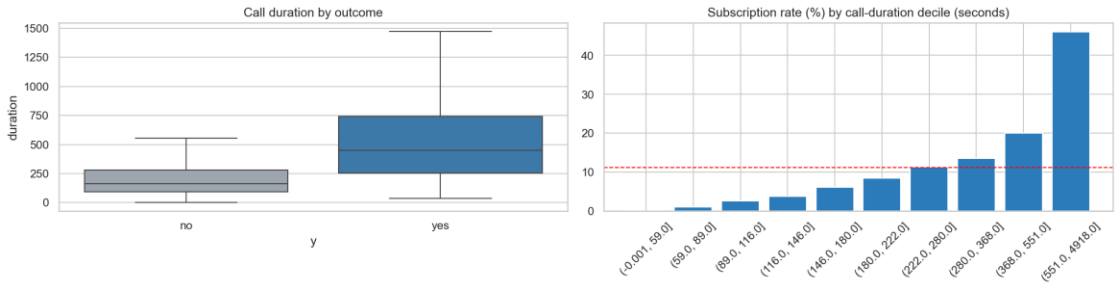


Figure 5 – Call duration vs outcome: evidence of leakage

3.3 EDA insight log

ID	Insight	Evidence	Implication for the task	Action
E1	Target imbalanced: 11.27% yes (7.9 : 1)	1.7	Accuracy misleading ("always no" = 88.73%)	PR-AUC, recall/F1, stratify, imbalance handling (Parts 2–3)
E2	One row = one client contact; no client ID; date-ordered 2008–2010	1.6	Repeat clients cannot be detected	Limitation
E3	12 exact duplicates, 0 label conflicts	1.6	Could appear in both train and test	Remove before split (D8)

ID	Insight	Evidence	Implication for the task	Action
E4	26.0% of rows contain "unknown"; unknown `default` 5.15% vs 12.88% yes	1.11.1	"unknown" is informative	Keep as category (D7)
E5	`default = yes` only 3 rows	1.11.1, 1.11.3	Too rare to learn	Merge (D13)
E6	`housing`/`loan` unknown on the same 990 rows	1.11.1	Shared missingness	Note
E7	`pdays = 999` in 96.3%; 4,110 rows have 999 but `previous > 0`	1.11.2	Raw `pdays` unreliable	`contacted_before` flag (D10)
E8	Outliers are real and predictive (age > 69.5: 47.12% yes)	1.8	Removing them deletes signal	Keep; cap `campaign` (D9)
E9	`poutcome = success` → 65.11% yes; largest chi-square (4,230.5)	1.10.1–1.10.2	Key predictor and profile	Keep; profiles (Part 6)
E10	Economic indicators are the strongest numeric inputs (AUC 0.72–0.75)	1.10.3	Important inputs	Keep; handle collinearity
E11	Mar/Sep/Oct/Dec at 44–51% but few rows; May 6.43% with the most calls	1.10.1	Strong but partly a time proxy	Interpret with care
E12	Age is U-shaped (41–50: 8.17%; 71+: 47.87%); linear AUC 0.511	1.10.4	Linear models miss it	Age groups for LR (2.5)
E13	Collinear economic indicators (VIF 64.3 / 33.1 / 31.6)	1.12	Unstable LR coefficients	Keep one for LR (D11)
E14	Yes-rate 4.84% → 19.48% → 52.14% across 2008–2010	1.13	Random split may be optimistic	Time-drift check (D12)
E15	Contact fatigue: 13.04% at 1 call → 3.56% at 10+	1.10.4	Diminishing returns	Cap calls (recommendation)
E16	Mobile 14.74% vs landline 5.23%; students 31.43%, retired 25.23%	1.10.1	Useful profile features	Keep; profiles
E17	`housing` (p = 0.058), `loan` (p = 0.58) not significant; `day_of_week` weak	1.10.2	Low predictive value	Feature selection (§2.3)
E18	`duration` AUC 0.818 > any pre-call input	1.14	Post-call leakage	Exclude (D3)
E19	`campaign` counts calls including the last one	1.5	Only partly known before contact	Keep and justify (§2.1f)

4. Preprocessing and Feature Engineering

Preprocessing was split into two kinds to prevent leakage. **Row-level cleaning** uses fixed rules that learn nothing from the data and is applied before the split. **Fitted transformations** (one-hot encoding, scaling) learn from data, so they are placed inside a scikit-learn `Pipeline` and fitted on training data only — re-fitted on every cross-validation fold.

4.1 Steps

- **Cleaning:** remove 12 duplicates (41,188 → 41,176 rows); drop `duration`; replace `pdays` with a `contacted_before` flag (it matches `poutcome` exactly); merge `default = yes` (3 rows) into `not_no` and `illiterate` (18 rows) into `basic.4y`; cap `campaign` at 10 (869 rows).
- **Split before any data-driven decision:** stratified 80/20 split with seed 42 – train 32,940 rows, test 8,236 rows, both exactly 11.27% yes. The test set stays locked until evaluation.
- **Feature selection on training data:** `housing` (χ^2 2.7, p = 0.26) and `loan` (χ^2 0.2, p = 0.92) are unrelated to the target and dropping them changes CV PR-AUC by less than 0.001, so they were removed; `day_of_week` was kept because removing it lowered PR-AUC in both test models.
- **Collinearity fix for Logistic Regression:** keep only `euribor3m` of the three correlated indicators – maximum VIF 64.6 → 4.7. Tree models keep all five.
- **Two pipelines:** linear (one-hot + age groups for the U-shape + standard scaling; 58 columns) and tree (one-hot + raw numbers; 54 columns).

4.2 Class imbalance

The training data is 7.9 : 1 (29,229 no vs 3,711 yes). It was handled in layers: (1) PR-AUC and top-20% capture instead of accuracy; (2) stratified split and folds; (3) **class weights** in all four models (`class_weight="balanced"`: no = 0.563, yes = 4.438; XGBoost `scale_pos_weight` = 7.88), which change no data and carry no leakage risk; (4) SMOTE was tested inside the CV folds but reduced PR-AUC for every tree model (Section 6.4), so it was rejected; (5) the decision threshold was tuned on training out-of-fold predictions (Section 7.3).

4.3 Additional check: removing outliers

To confirm decision D9 (keep outliers), the tuned Random Forest and Logistic Regression were re-trained after removing IQR outliers from the training data and scored on the same test set (separate check, not part of the notebook):

Variant	Train rows	RF accuracy	RF PR-AUC	RF top-20%	LR accuracy	LR PR-AUC
A. Keep all outliers (chosen)	32,940	**0.864**	**0.496**	**67.1%**	**0.816**	**0.446**
B. Remove outliers in age, campaign, cons.conf.idx	30,347	0.864	0.487	67.1%	0.808	0.440
C. Remove all IQR outliers incl. previous	26,209	0.840	0.440	65.6%	0.736	0.381

Variant	Train rows	RF accuracy	RF PR-AUC	RF top-20%	LR accuracy	LR PR-AUC
D. Remove outliers from train and test (not valid)	26,209	0.868	0.354	58.0%	0.780	0.296

Removing outliers never helped and removing all of them lowered Random Forest PR-AUC from 0.496 to 0.440, because the IQR rule flags every previously contacted client (26.65% yes) as an outlier. Variant D only *looks* more accurate because deleting outliers removes many subscribers from the test set (yes-rate 8.66%, where "always no" scores 91.3%); it is not valid, since real clients cannot be deleted from a call list.

4.4 Preprocessing / feature log

#	Issue	Options	Action	Justification	Effect / evidence
P1	12 duplicates (E3)	keep; remove	Removed	Avoid identical rows in train and test	41,188 → 41,176 (2.1a)
P2	`duration` leakage (E18)	keep; drop	Dropped (benchmark in §3.3)	Not known before the call	2.1b, 3.3
P3	`pdays` placeholder (E7)	raw; impute; flag	`contacted_before` flag	Consistent with `poutcome`	8.83% vs 26.65% yes (2.1c)
P4	`default = yes` × 3 (E5)	keep; merge	`no` / `not_no`	Too rare	32,577 / 8,599 (2.1d)
P5	`illiterate` × 18	keep; merge	Merged into `basic.4y`	Too rare	2.1e
P6	`education` encoding	ordinal; one-hot	One-hot	Rates not ordered; "unknown" has no order	1.10.1
P7	"unknown" in 26% of rows (E4)	delete; impute; category	Kept as category	Informative	No rows lost
P8	Outliers (E8)	remove; cap; keep	Kept; `campaign` capped at 10	Real and predictive	869 rows capped (2.1f)
P9	Hold-out split	random; time-based	Stratified 80/20, seed 42, before selection	Decisions must use train only	32,940 / 8,236 (2.2)
P10	Weak features (E17)	keep; drop	Drop `housing`, `loan`; keep `day_of_week`	Not significant; no CV loss	2.3
P11	Collinearity (E13)	keep; drop; regularise	LR keeps `euribor3m` only	VIF > 30	Max VIF 64.6 → 4.7 (2.4)
P12	Non-linear age (E12)	raw; bins	LR: 7 age groups; trees: raw age	LR cannot learn a U-shape	2.5
P13	Encoding / scaling	—	One-hot (`handle_unknown="ignore"`); StandardScaler for LR only	Trees are scale-free	58 / 54 columns (2.5)
P14	Imbalance 7.88 : 1 (E1)	none; undersample; SMOTE; weights	Class weights; SMOTE compared (§4.7)	No synthetic data, no leakage risk	weights 0.563 / 4.438 (2.6)

5. Validation Design

How models would be judged was fixed before any model was built.

Metric	What it measures	Role
PR-AUC (average precision)	How well subscribers are ranked at the top; random ≈ 0.113	Primary metric
ROC-AUC	Ranking quality over both classes; random = 0.5	Secondary; compares periods
Recall / precision / F1 (yes)	Subscribers found / calls that succeed / balance	Business view
Balanced accuracy	Average recall of both classes	Secondary
Top-20% capture	Share of subscribers reached by calling the top-scored 20%; random = 20%	Business metric
Accuracy	Share of correct predictions	Reported only – misleading under imbalance

- **Scheme:** stratified 5-fold cross-validation (shuffled, seed 42) on the training set, the same folds for every model; each fold has 6,588 validation rows with about 742 subscribers.
- **Reference floor:** "always no" – accuracy 0.887 but recall 0 and PR-AUC 0.113; random guessing – PR-AUC 0.113, top-20% capture 0.199. Untuned models already reach PR-AUC 0.437 (LR) and 0.465 (RF).
- **Leakage benchmark:** adding `duration` would raise PR-AUC from 0.437 to 0.584 (LR) and from 0.465 to 0.656 (RF) – performance that is impossible before a call.
- **Time drift:** forward-in-time validation (train on the past, validate on later rows) gave mean ROC-AUC of only 0.515 (LR) and 0.550 (RF) vs 0.788 / 0.796 with shuffled CV. Causes: almost no contact history in 2008 (2.7% contacted before vs 59.0% in 2010), months unseen in training and interest rates outside the training range. Shuffled CV is kept to compare models; this is reported as a key limitation (D12).

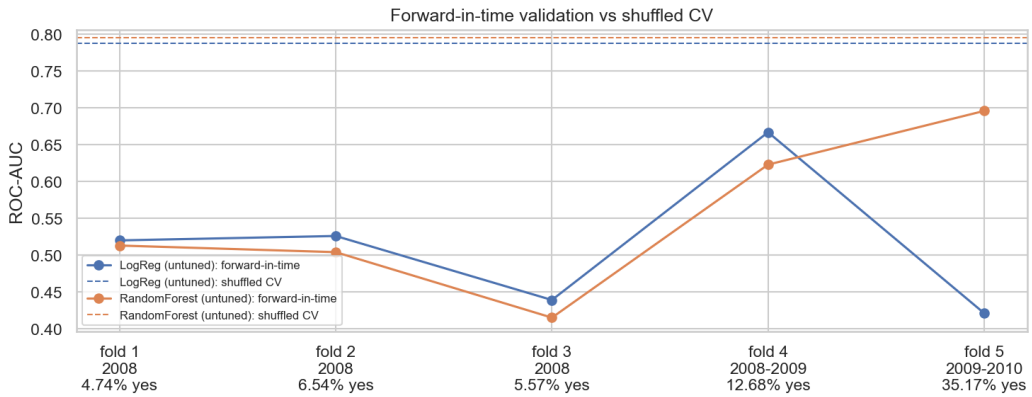


Figure 6 –Forward-in-time validation vs shuffled cross-validation

6. Model / Method Comparison

6.1 Models and tuning

Four models were built as full pipelines (preprocessing + model), tuned on the training set with the same 5 folds, scored by PR-AUC and using class weights.

Model	Role	Why	Search	Best settings	CV PR-AUC
Logistic Regression	Baseline	Simple, interpretable	Grid, 6	C = 0.01	0.4376 ± 0.0125
Decision Tree	Alternative 1	Non-linear, readable rules	Grid, 80	entropy, no depth limit, min 100 per leaf	0.4433 ± 0.0156
Random Forest	Alternative 2	Many de-correlated trees, robust	Random, 20	300 trees, depth 12, min 10 per leaf, 30% features	0.4702 ± 0.0176
XGBoost	Alternative 3	Gradient boosting, strong on tabular data	Random, 25	600 rounds, lr 0.01, depth 5, subsample 0.7	0.4674 ± 0.0166

6.2 Cross-validated comparison

Model	Role	PR-AUC	ROC-AUC	Recall	Precision	F1	Top-20%	Folds better than LR	Train–CV gap	Interpretability
Dummy: always no	floor	0.113	0.500	0.000	0.000	0.000	0.189	–	–	–
Logistic Regression	baseline	0.438	0.787	0.634	0.328	0.432	0.620	–	0.5%	High (odds ratios)
Decision Tree	alternative 1	0.443	0.783	0.652	0.315	0.425	0.625	3/5	7.9%	High (rules)
Random Forest	alternative 2	**0.470**	**0.801**	0.603	**0.411**	**0.489**	**0.643**	**5/5**	25.5%	Medium (feature importance)
XGBoost	alternative 3	0.467	0.800	0.631	0.372	0.468	0.642	**5/5**	14.0%	Medium (feature importance)

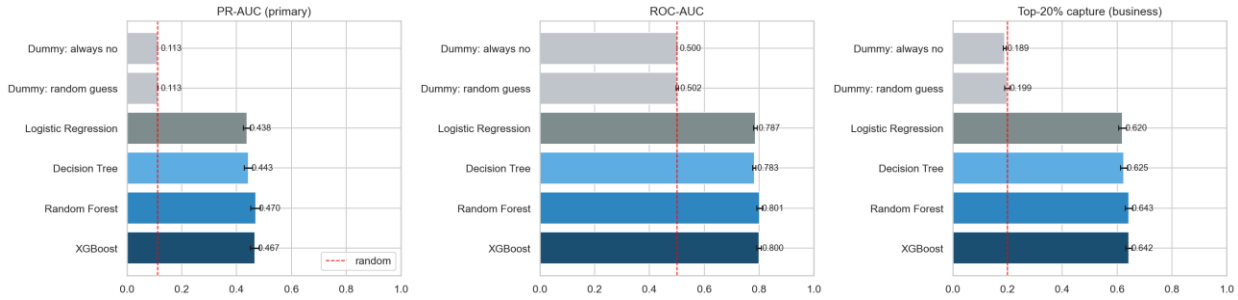


Figure 7 – Cross-validated PR-AUC, ROC-AUC and top-20% capture (red line = random)

- Random Forest (+0.033) and XGBoost (+0.030) beat the baseline in **all 5 folds**; the Decision Tree only by +0.006 in 3 of 5 folds.
- Random Forest and XGBoost differ by 0.003, far below the fold-to-fold standard deviation (≈ 0.017), so they were shortlisted together (D21).

- Tuning gains were small and the best settings of each model lie close together, suggesting that performance is limited by the information available before a call.

6.3 Overfitting

Model	Train PR-AUC	CV PR-AUC	Gap	Gap % of CV
Logistic Regression	0.4399	0.4376	0.0023	0.5%
Decision Tree	0.4784	0.4433	0.0351	7.9%
Random Forest	0.5900	0.4702	0.1198	25.5%
XGBoost	0.5327	0.4674	0.0653	14.0%

6.4 Imbalance method: class weights vs SMOTE

Model	PR-AUC class weights	PR-AUC SMOTE	Change
Logistic Regression	0.4376	0.4378	+0.0002
Decision Tree	0.4433	0.3815	−0.0618
Random Forest	0.4702	0.4433	−0.0269
XGBoost	0.4674	0.4481	−0.0193

SMOTE never improved ranking quality and hurt every tree model (it interpolates one-hot columns into unrealistic "half-category" clients), so class weights were kept (D16).

7. Evaluation on the Test Set

7.1 Test results

Model	PR-AUC	95% CI (bootstrap)	ROC-AUC	Accuracy	Balanced acc.	Top-20% capture
Dummy: always no	0.113	–	0.500	0.887	0.500	19.4%
Logistic Regression	0.446	0.413 – 0.482	0.797	0.816	0.749	64.3%
Decision Tree	0.465	0.431 – 0.499	0.792	0.788	0.741	64.2%
Random Forest	**0.496**	0.461 – 0.530	**0.814**	**0.864**	**0.765**	**67.1%**
XGBoost	0.484	0.449 – 0.520	0.814	0.846	0.763	66.5%

- Every test score lies within two CV standard deviations of its CV estimate, so the models **generalise** and nothing was over-tuned.
- Bootstrap (1,000 resamples): Random Forest − XGBoost = **+0.011** (95% CI 0.003 – 0.019, RF better in 99.8% of resamples); Random Forest − Logistic Regression = +0.049 (0.033 – 0.066).
- Accuracy is shown for completeness: "always no" is the most accurate (0.887) yet useless. Among real models Random Forest is also the most accurate (0.864).

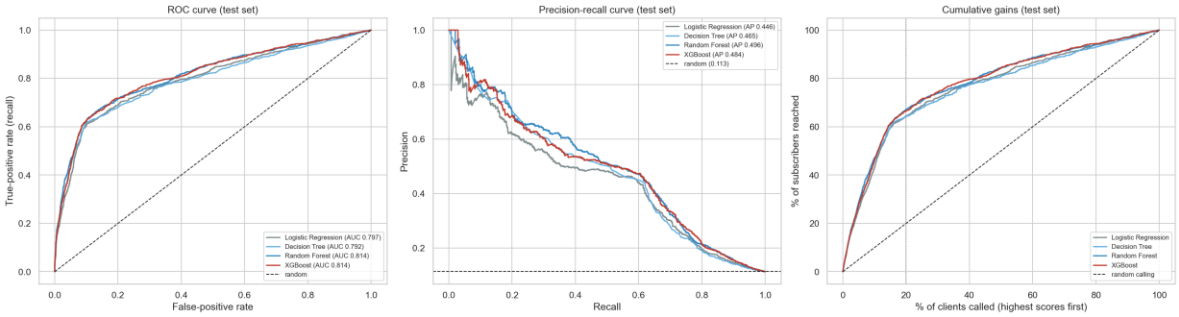


Figure 8 – ROC, precision–recall and cumulative gains on the test set

7.2 Business view: gains

Calling the top 10% of the Random Forest ranking reaches 47.2% of all subscribers (4.7× random calling), the top 20% reaches 67.1% and the top 50% reaches 84.4%. The gains curve rises steeply up to about 15–20% of clients and then flattens.

7.3 Decision thresholds

Thresholds were chosen from **out-of-fold predictions on the training set** (never from the test set) for three business rules, then applied unchanged to the test set:

Random Forest rule	Threshold	% called	Calls	Subscribers found	Conversion	vs call everyone
Max F1	0.579	14.6%	1,206	563 (60.7%)	46.7%	4.1×
Capacity 20%	0.424	20.1%	1,653	623 (67.1%)	37.7%	3.3×
Recall ≥ 70%	0.364	28.0%	2,307	679 (73.2%)	29.4%	2.6×
Call everyone	–	100%	8,236	928 (100%)	11.3%	1×

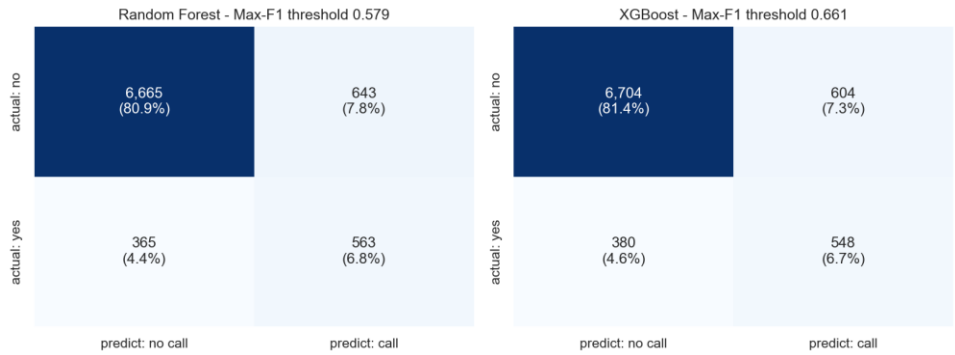


Figure 9 – Confusion matrices at the Max-F1 threshold

7.4 Results per year

Year	Test rows	Yes-rate	LR	DT	RF	XGB
2008	5,566	4.49%	0.560	0.537	**0.588**	**0.589**
2009	2,277	20.60%	0.755	0.770	**0.790**	0.784
2010	393	53.18%	0.688	0.742	**0.744**	0.730

Within-year ROC-AUC

Ranking within a single year is weaker than the overall ROC-AUC of 0.81 – part of the models' skill is separating good periods from bad ones. 2008, with almost no contact history, is the hardest year. Random Forest is best or tied-best in every year.

8. Interpretation and Marketing Profile Analysis

8.1 What drives the predictions

- **Permutation importance** (drop in test PR-AUC when a feature is shuffled): for Random Forest the top inputs are `nr.employed` (0.085), `poutcome` (0.059), `euribor3m` (0.059), `emp.var.rate` (0.040) and `contact` (0.035). Individual personal attributes (age, job, education, marital) each cost less than 0.005.
- **Random Forest feature importance** (tree-based): the five economic indicators account for **62.1%** of the forest's importance. It agrees with permutation importance on the main drivers (Spearman 0.72) but over-rates features with many distinct values such as age (4th vs 11th), so permutation importance is the more reliable guide.
- Random Forest and XGBoost agree almost perfectly on feature ranking (Spearman 0.96).
- **Logistic Regression odds ratios:** previous success × 2.04; each +1 SD of Euribor × 0.42; May × 0.50; previous failure × 0.50; March × 1.83; mobile × 1.21; retired × 1.21; students × 1.17.

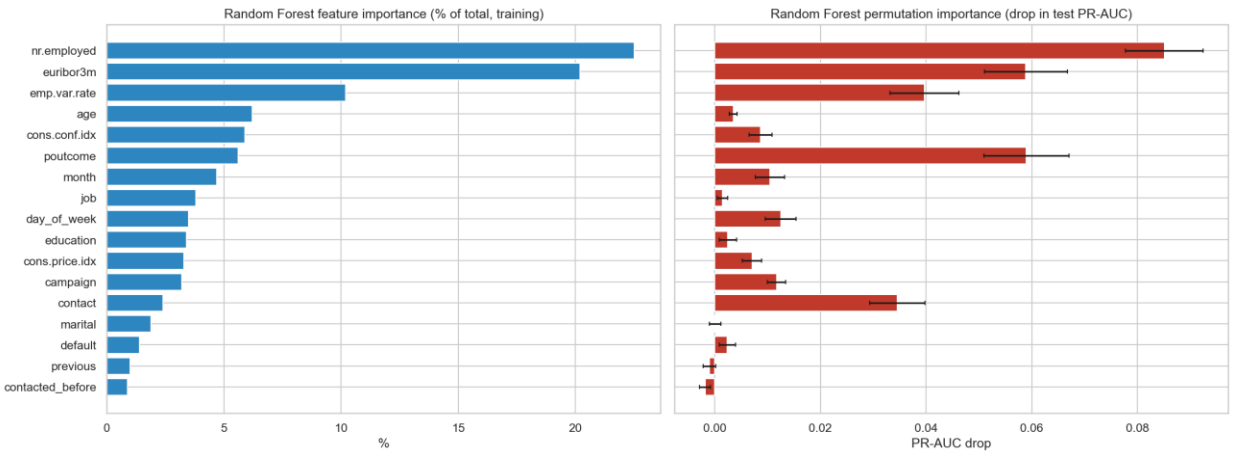


Figure 10 – Random Forest feature importance vs permutation importance

Interpretation: the model learns mostly *when* a campaign works (economic conditions), then *who* responds (contact history and channel). This explains the time-drift limitation and supports the recommendation to retrain regularly.

8.2 Marketing profiles (secondary lens)

Profile	Evidence	Yes-rate (test)	Lift
Previous success, contacted more than once before	Client segment S1 (6.4)	72.0%	6.4×
Previous success, first call in this campaign	Client segment S2 (6.4)	65.9%	5.9×
Previous success in a low-employment period	All-feature segment S1 (6.4)	76.7%	6.8×
Mobile clients over 60	Client segment S4 (6.4)	41.2%	3.7×
Students and retired clients	1.10.1 (31.43% / 25.23%); top-10% over-representation 5.8× / 4.4× (6.5); LR odds ×1.17 / ×1.21 (6.3)	–	2.2–2.8×
Landline clients under 58, no previous success (largest low group)	Client segment S7 (6.4)	4.8%	0.43×
Normal-economy periods (high employment)	All-feature segments S7–S8 (6.4)	3.5–6.5%	0.3–0.6×
Called 5+ times in the same campaign	1.10.4 (≤ 7.5%)	–	< 0.7×
Default status not confirmed "no"	1.11.1 (5.15%); LR odds ×0.89 (6.3)	–	0.46×

The top 10% of clients ranked by Random Forest subscribe at 53.2% vs 6.6% for the rest; they are mostly contacted in low-interest periods (median Euribor 0.88 vs 4.86), 42.8% were contacted before (vs 10.3%) and 92.7% were reached by mobile.

9. Final Model, Recommendations and Limitations

9.1 Final model (D27, D28)

Item	Value
Model	Random Forest: 300 trees, max_depth 12, min_samples_leaf 10, max_features 0.3, balanced class weights
Inputs (15)	job, education, default, contact, month, day_of_week, poutcome, campaign, previous, contacted_before, emp.var.rate, cons.price.idx, cons.conf.idx, euribor3m, nr.employed
Excluded	`duration` (leakage), `housing`/`loan` (no signal), `age`/`marital` (data minimisation), raw `pdays`
Test performance	PR-AUC 0.496 · ROC-AUC 0.814 · top-20% capture 67.0%
Main rule	Capacity 20%: call the top 20% of the ranked list (622 of 928 subscribers, conversion 37.7%)
Alternative rule	Max F1 (threshold 0.638): conversion 47.6% with 13.7% of the list called
Fallback model	XGBoost (test PR-AUC 0.484)

Verified on the test set (trained on the training set only): PR-AUC 0.496, ROC-AUC 0.814, top-20% capture 67.0%. Capacity 20%: 1,651 calls → 622 subscribers, conversion 37.7%. Max F1 (threshold 0.638): 1,125 calls → 536 subscribers, conversion 47.6%. Within-year ROC-AUC 0.592 / 0.788 / 0.745.

9.2 Stakeholder value

	Call everyone	Model: call top 20% (main)	Model: Max F1 (alternative)
Calls	8,236	**1,651 (−80%)**	1,125 (−86%)
Subscribers reached	928	**622 (67%)**	536 (58%)
Conversion per call	11.3%	**37.7% (3.3×)**	47.6% (4.2×)
Wasted calls	7,308	**1,029**	589

The bank can reach two thirds of its subscribers with one fifth of the calls, saving about 6,600 calls per 8,000 clients and sparing uninterested clients an unwanted call.

9.3 Recommendations

#	Recommendation	Evidence
R1	Rank every campaign list with the model and **call the top 20% first**; extend down the list only if capacity remains	7.4; gains curve flattens after ~15–20% (§5.3)
R2	Use the **Max F1 rule** when staff is short or calls are expensive	§7.4
R3	**Prioritise previous subscribers** and keep complete contact-history records	65–77% yes (§6.4); weakest ranking in 2008 without history (§5.6)
R4	**Prefer mobile numbers**; try to obtain mobile numbers for landline-only clients	14.74% vs 5.23% (§1.10.1); landline < 58: 4.8% (§6.4)
R5	**Cap calls at about 4 per client per campaign**	13.04% at 1 call → 3.56% at 10+ (§1.10.4)
R6	Plan campaign intensity around economic conditions; review the mass-calling strategy in May	Economic indicators dominate importance (§6.1–6.2); May 6.43% (§1.10.1)
R7	**Retrain before each campaign** (at least quarterly) and monitor conversion of the called list	Forward-in-time ROC-AUC ≈ 0.55 (§3.4)
R8	**Human review and fairness monitoring**: track selection by age and marital group; extra care with older clients	61+ selected at 95–99% even without age (§6.6, §7.4)
R9	Improve data: stable client ID, consistent contact history, account balance	No client ID; pdays inconsistent in 4,110 rows (§1.11.2)
R10	Use the marketing profiles (§6.8) to design messages for high-response groups	Part 6

9.4 Limitations

#	Limitation	Mitigation
L1	**Time drift:** past → future validation is much weaker (ROC-AUC ≈ 0.55)	Frequent retraining, monitoring (R7)
L2	Random split from the same period, so the test score (0.496) is likely optimistic for future campaigns	Reported with within-year (0.59–0.79) and forward-in-time results
L3	Old, single-bank data (Portugal, 2008–2010 crisis)	Retrain on the bank's own recent data
L4	Economic features dominate, so the model partly predicts *when* rather than *who*	Prefer history/channel profiles (R3, R4)
L5	Performance ceiling without call duration (PR-AUC ≈ 0.50); a third of subscribers fall outside the top 20%	Extend the list when capacity allows
L6	Scores are not calibrated probabilities	Use as a ranking

#	Limitation	Mitigation
L7	No client ID; 26% "unknown" values; inconsistent pdays	Data improvements (R9)
L8	campaign only partly known before a campaign starts	Re-score before each call attempt
L9	Random Forest overfits the training data (gap 25.5%) without hurting test performance	Monitor; XGBoost fallback
L10	Proxies for sensitive attributes	Group monitoring and human oversight (R8)

9.5 Risks

- **Fairness:** older clients are selected far more often; this reflects their real response, but risks over-contacting potentially vulnerable customers. Use call caps, clear product information and monitoring.
- **Privacy:** only data the bank already holds; age and marital removed from the final model (data minimisation).
- **Automation:** the model supports call planning; it must not decide eligibility for products.
- **Feedback loop:** if only top-scored clients are called, the model stops learning about others, so keep a small random sample (e.g. 5%) of lower-ranked clients in each campaign.

10. Decision Log

Every important decision with the options considered, the choice, the reason, the evidence and the owner.

ID	Step	Decision	Options considered	Choice	Reason	Owner
D1	Part 1 Setup	Dataset version	bank-full.csv; bank.csv (10% sample); bank-additional-full.csv; bank-additional.csv (10% sample)	bank-additional-full.csv	Contains 5 economic indicators that the dataset authors report improve prediction even without call duration; they are known before a call. Full row count. Trade-off: no balance column	Karunathilaka S.H.T.R
D4	Part 1 Setup	Random seed	Unfixed; fixed	RANDOM_STATE = 42 for split, folds, searches, models and bootstrap	Reproducible results	Karunathilaka S.H.T.R
D30	Delivery	Division of work	Roles as in the initial report; one model per member	One model + one main stage per member (see title page)	Every member owns a model and a stage; even workload; clear individual contribution	All
D32	Part 1 / 2	Test for categorical association	Chi-square only; chi-square + Cramér's V	Chi-square test of independence (statistic, dof, p-value) with	Chi-square is the filter method from the lectures; rate tables	Karunathilaka S.H.T.R

				subscription-rate tables	show the size of each effect	
D3	Part 2 Preprocessing	Treatment of `duration`	Keep; drop; benchmark only	Drop from the realistic model; benchmark only	Not known before the call; alone it reaches AUC 0.818 and would add +0.15 / +0.19 PR-AUC	Jayasuriya J.A.Y.N
D7	Part 2 Preprocessing	"unknown" values	Delete rows; impute mode; own category	Keep as own category	26.0% of rows affected; unknown `default` 5.15% vs 12.88% yes, so informative	Jayasuriya Jayasuriya J.A.Y.N
D8	Part 2 Preprocessing	Duplicate rows	Keep; remove	Remove the 12 exact duplicates	Could appear in both train and test; no label conflicts	Jayasuriya J.A.Y.N
D9	Part 2 Preprocessing	Outliers	Remove; cap; keep	Keep all rows; cap `campaign` at 10	Values are real and predictive (age > 69.5: 47.12% yes); sparse campaign tail could distort LR	Jayasuriya J.A.Y.N
D10	Part 2 Preprocessing	`pdays` placeholder	Raw; NaN + impute; flag	`contacted_before` flag; drop `pdays`	96.3% placeholder; 4,110 contradictory rows; flag matches `poutcome` exactly	Jayasuriya J.A.Y.N
D11	Part 2 Preprocessing	Collinear economic indicators	Keep all; drop; regularise	LR keeps `euribor3m` (+ 2 other indicators); trees keep all 5	VIF up to 64.6 makes LR coefficients unstable; max VIF after = 4.7	Karunathilaka S.H.T.R
D13	Part 2 Preprocessing	`default` with 3 "yes" rows	Keep; drop rows; merge	`no` vs `not_no`	Too rare; both mean default-free status is not confirmed	Jayasuriya J.A.Y.N
D14	Part 2 Preprocessing	Rare `illiterate` (18) and encoding	Keep; merge; ordinal vs one-hot	Merge into `basic.4y`; one-hot	Too rare; rates not ordered by level	Jayasuriya J.A.Y.N
D15	Part 2 Preprocessing	Feature selection	Keep all; drop housing+loan; also drop day_of_week	Drop `housing`, `loan`; keep `day_of_week`	Training-set chi-square p = 0.26 / 0.92; dropping them changes CV PR-AUC by < 0.001; dropping day_of_week lowered PR-AUC in both models	Jayasuriya J.A.Y.N
D16	Part 2 / 4	Class-imbalance handling	None; undersampling; SMOTE; class weights; threshold	Class weights; SMOTE compared; threshold tuned in Part 5	No synthetic data or leakage risk; SMOTE never improved PR-AUC (−0.062 to +0.000)	Thulmanthi W.A.S
D17	Part 2 Preprocessing	When to split	Later; before feature selection	Stratified 80/20, seed 42, before feature selection and VIF	Data-driven decisions must use training data only	Jayasuriya J.A.Y.N

D12	Part 3 Validation	Validation under time drift	Shuffled stratified CV; forward-in-time CV	Shuffled CV for comparing models; forward-in-time reported as a limitation; per-year check in Part 5	Forward-in-time ROC-AUC 0.515 / 0.550 vs 0.788 / 0.796 shuffled; 2008 had almost no contact history	Thennakoon T.M.S.D
D18	Part 3 Validation	Validation scheme and metrics	Accuracy; ROC-AUC; PR-AUC; F1; business metric	Stratified 5-fold CV, same folds; PR-AUC primary; top-20% capture business metric	"Always no" has the highest accuracy (0.887) but recall 0	Thennakoon T.M.S.D
D2	Part 4 Modelling	Baseline model	Dummy classifier; Logistic Regression	Logistic Regression baseline + 3 alternatives; dummies as reference rows	Rubric asks for a sensible baseline; LR is simple and interpretable	Karunathilaka S.H.T.R
D5	Part 4 Modelling	Boosting library	XGBoost; LightGBM; sklearn GradientBoosting	XGBoost	Strong on tabular data; `scale_pos_weight` for imbalance	Thennakoon T.M.S.D
D19	Part 4 Modelling	Tuning approach	Manual; full grid; randomised search	Grid for LR (6) and DT (80); randomised for RF (20) and XGBoost (25); same folds; PR-AUC	Small spaces fit a grid; large spaces use randomised search with a fixed seed	Thulmanthi W.A.S
D21	Part 4 Modelling	Shortlist	All four; best only; best two	Random Forest and XGBoost	Both beat the baseline in 5/5 folds (+0.033 / +0.030) and tie with each other; DT only +0.006 in 3/5	Thennakoon T.M.S.D
D22	Part 5 Evaluation	Choosing the decision threshold	Default 0.5; tune on test; tune on training OOF	Training OOF with three business rules, applied unchanged to test	Tuning on the test set would leak; capacity rule called 20.1% of test clients	Thennakoon T.M.S.D
D23	Part 5 Evaluation	Statistical comparison on test	Single score; bootstrap CIs	1,000-resample bootstrap CIs	RF − XGB = +0.011 [0.003, 0.019]; RF − LR = +0.049 [0.033, 0.066]	Thennakoon T.M.S.D
D24	Part 6 Interpretation	Interpretation methods	Permutation importance; tree-based importance; LR coefficients; rule tree	Permutation importance (LR, RF, XGB) + RF feature importance + LR odds ratios + 3-level rule tree	Methods from the lectures that cross-check each other; RF–XGB ranking agreement 0.96	Thulmanthi W.A.S
D25	Part 6 Interpretation	Segment rules: which features	All features; client + history only	Report both	The all-feature tree splits first on the economy (time periods); client-only	Jayasuriya J.A.Y.N

					rules give customer types	
D26	Part 6 / 7	Sensitive attributes `age`, `marital`	Keep; remove	Remove from the final model; monitor group outcomes	No performance cost (0.4959 vs 0.4955) but proxies keep selection of 61+ at 95–99%	Jayasuriya J.A.Y.N
D27	Part 7 Final model	Final model	LR; DT; RF; XGBoost	Random Forest (tuned, without age & marital); XGBoost fallback	Best test PR-AUC 0.496 (> XGB +0.011, > LR +0.049), best top-20% capture, best within-year ranking	Thulmanthi W.A.S
D28	Part 7 Final model	Operating rule	Default 0.5; Capacity 20%; Max F1; Recall ≥ 70%	Capacity 20% (main); Max F1 (alternative, threshold 0.638)	Capacity matches call-centre planning (622 subscribers, 37.7%); Max F1 maximises efficiency (47.6%)	Thennakoon T.M.S.D

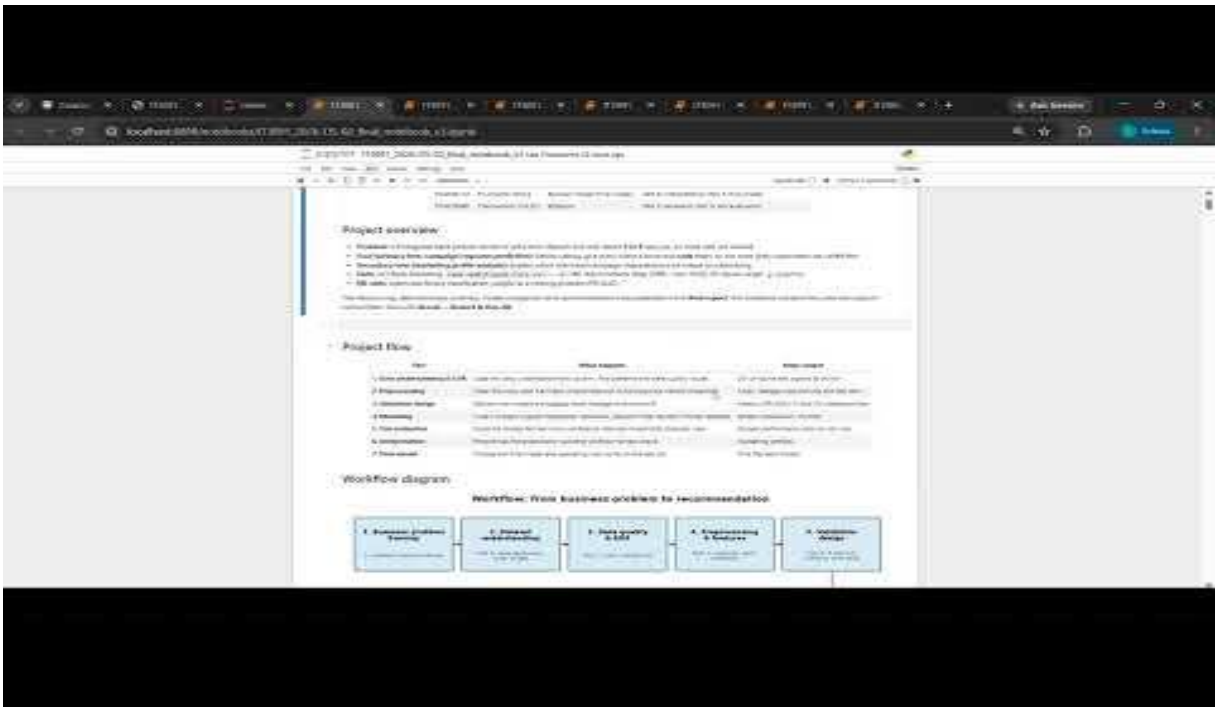
11. Conclusion

The project built a model that ranks bank clients before a term-deposit call using only information available before the call. After careful handling of leakage (`duration`), data quality and class imbalance, four models were compared on equal terms. **Random Forest** was the best (test PR-AUC 0.496 vs 0.446 for the Logistic Regression baseline and 0.113 for random guessing). Calling the top 20% of its ranking makes 80% fewer calls, still reaches two thirds of subscribers, and raises the success rate per call from 11% to 38%.

The analysis also showed the model's limits honestly: its skill depends strongly on economic conditions and contact history, so it must be retrained before every campaign, and older clients are selected far more often even without age as an input, so group outcomes must be monitored.

Demo video link

https://youtu.be/gqpUobX0B-c



Appendix A – Reproducibility

Item	Detail
Final notebook	`notebooks/IT3091_2026-DS-02_final_notebook_v3.ipynb` – self-contained; run with Kernel → Restart & Run All
Data	`data/raw/bank-additional-full.csv` (41,188 × 21, separator `;`)
Environment	Python 3.13.7, pandas 2.3.2, scikit-learn 1.7.1, XGBoost 3.2.0, imbalanced-learn 0.14.1
Randomness	`random_state = 42` for the split, CV folds, searches, models and bootstrap – results are repeatable
Runtime	About 14 minutes
Figures	Saved by the notebook into `figures/`

Appendix B – AI-Use Declaration

What the AI was used for:

Used Claude AI

Activity	How the AI was used
Understanding the brief	Summarized the assignment descriptor and Concepts of applied theories and models.
Code	Assistant for write code while directed by each member.
Documentation	Drafted the decision logs, under direction of members, used as a writing assistant.
Explanations	Explained concepts (PR-AUC, leakage, SMOTE, thresholds, proxies) in full details on request.